ZORAASI RESEARCH PROGRAM / 11 OCTOBER 2026

Quadratic Higgs-portal interactions and a conditional measurement bridge

Thermal scalar scattering, mixed fields, and visibility decay

Prepared for Christopher Michael Baird

AI-assisted derivation and computational audit; working note v0.1

Result. A massive free scalar has no zero-frequency noise in its *linear* field operator, but its quadratic operator has a nonzero thermal elastic-scattering contribution. The symmetric Higgs portal supplies just such a quadratic matter coupling. Under a declared thermal state, heavy fixed-path probe, and weak-coupling Markov approximation, this yields a calculable visibility-decay rate. The derivation extends to two mixed scalar mass modes.

Scientific status. This is a conditional application of established Higgs-portal and open-system methods, not a claim of new fundamental physics, experimental detection, novelty, or a completed Theory of Everything. The existence, population, and temperature of the putative scalar bath have not been measured. A complete laboratory model and high-energy completion remain open. The parent goal remains completion of the full theory; this note addresses one necessary physical interface without redefining that goal.

What this changes in the research record

Prior unresolved issue	Result supplied here
Scalar action to probe coupling	Tree-level matching gives a definite quadratic matter operator, including its sign and identical-scalar factor.
Massive-bath obstruction	The linear vacuum/DC obstruction survives; quadratic thermal elastic scattering opens a different channel.
Arbitrary visibility coefficient	A conditional rate follows from masses, portal matrix, bath temperature, and path separation. No exponential field gate is assumed.
Two-scalar mixing	An ordered-pair spectral sum includes both diagonal scattering and mixed-species conversion.
Common-mode null	Equal means cancel phase, but do not generally cancel fluctuating path information.
Distance law	Quadratic growth applies at small separation; single-scalar rates saturate at large separation.

Units and notation. Throughout, $\hbar = c = k_B = 1$; T is bath temperature, τ is interrogation time, r is path separation, m is a scalar mass, and m_h is the Higgs mass. Temperature is not interrogation time. To convert the rate to inverse seconds divide its energy value by $\hbar$; physical distance enters as $r/(\hbar c)$ in inverse-energy units. No numerical laboratory parameters are assigned.

1 From the stated portal to matter

Use metric signature $(+---)$ and the potential convention of the September two-singlet audit in the supplied corpus [1]:

$$V \supset \tfrac{1}{2}\kappa\phi^2 H^\dagger H, \qquad H = \begin{pmatrix} 0 \\ (v+h)/\sqrt{2} \end{pmatrix}. \tag{1}$$

For an elementary fermion, the terms linear in the physical unmixed Higgs are

$$\mathcal{L} \supset -\tfrac{1}{2}h(\Box + m_h^2)h - hJ_h, \qquad J_h = \tfrac{1}{2}\kappa v\phi^2 + \frac{m_f}{v}\bar{f}f. \tag{2}$$

Completing the square, at momenta small compared with m_h,

$$\Delta\mathcal{L} = \tfrac{1}{2}J_h(\Box + m_h^2)^{-1}J_h \supset c_f\phi^2\bar{f}f, \qquad c_f = \frac{\kappa m_f}{2m_h^2}. \tag{3}$$

For a nucleon define f_N by $\mathcal{L}_{hNN} = -(f_N m_N/v)h\bar{N}N$. Then $c_N = \kappa f_N m_N/(2m_h^2)$. In the nonrelativistic limit,

$$H_{\rm int} = -c_a \int \mathrm{d}^3x\, n_a(x)[\phi^2(x)]_R. \tag{4}$$

The positive Lagrangian coefficient becomes a negative potential-energy coefficient. Rates below depend on c_a^2, so this sign does not affect them. The $\phi\phi\bar{f}f$ vertex is $i\,2c_f$, not ic_f.

Scattering normalization. For scalar–nucleon scattering at low velocity, Eq. (3) gives

$$\sigma_{\phi N} = \frac{\kappa^2 f_N^2 m_N^4}{4\pi m_h^4 (m+m_N)^2} \longrightarrow \frac{c_N^2}{\pi} \quad (m \ll m_N). \tag{5}$$

This agrees with Eq. 8 of Athron et al. [2] in the same portal normalization. For a composite atom or extended object, its response and form factor must be derived; a nucleon coefficient must not be silently promoted to an optical or macroscopic detector coefficient.

Two mixed scalars

For a real interaction-basis vector φ and real symmetric portal matrix $K = K^T$, write $V \supset \tfrac{1}{2}\varphi^T K\varphi H^\dagger H$. An orthogonal rotation $\varphi = Rs$ diagonalizes the physical singlet mass matrix. With an unmixed Higgs and exact symmetric vacuum, the probe sees

$$B = s^T C s, \qquad C = \frac{f_a m_a}{2m_h^2} R^T K R, \qquad H_{\rm int} = -P_L B_L - P_R B_R. \tag{6}$$

Here $f_a = 1$ for an elementary fermion in the matching approximation; for nucleons it is defined above. All scalar masses must be positive for the massive equilibrium branch used here. Higgs–singlet mixing, appreciable self-interactions, and additional mediators require new matching. Off-diagonal C preserves simultaneous inversion $s \mapsto -s$; separate sign flips of individual mass modes need not be symmetries.

Source constraint. The matter term yields $(\Box + m^2)\phi = 2c_a n_a\phi + \cdots$, or its matrix counterpart $(\Box I + M^2)s = 2n_a C s + \cdots$. It changes the local mass matrix; it does not provide an additive source independent of the field. Classical zero field remains a solution. A thermal population is additional state information, not a consequence of this matching.

2 The thermal quadratic spectrum

For one free real scalar in homogeneous equilibrium at zero chemical potential, define

$$\omega_k = \sqrt{k^2 + m^2}, \quad n(\epsilon) = \frac{1}{e^{\epsilon/T} - 1}, \quad \mathrm{d}\mu_k = \frac{\mathrm{d}^3 k}{(2\pi)^3\, 2\omega_k}. \tag{7}$$

Let $O =: \phi^2 :$ and center it by subtracting its thermal mean. Vacuum normal ordering is the convention at this free-field order. Wick contraction gives, away from coincident-distribution ambiguities,

$$G^{>}(t, \mathbf{r}) = \int \mathrm{d}\mu_k \big[(1 + n_k) e^{-i\omega_k t + i\mathbf{k}\cdot\mathbf{r}} + n_k e^{i\omega_k t - i\mathbf{k}\cdot\mathbf{r}}\big], \tag{8}$$

$$C_O^{>}(t, \mathbf{r}) = \langle \delta O(t, \mathbf{r}) \delta O(0, 0) \rangle = 2[G^{>}(t, \mathbf{r})]^2. \tag{9}$$

Define a two-sided spectrum by $S_O(\Omega, r) = \int \mathrm{d}t\, e^{i\Omega t} C_O^{>}(t, \mathbf{r})$. At zero frequency the pair terms cannot contribute: $\omega_k + \omega_p \geq 2m$. Elastic thermal terms do contribute:

$$S_O(0, r) = 8\pi \int \mathrm{d}\mu_k \mathrm{d}\mu_p\, n_k (1 + n_p) \delta(\omega_k - \omega_p) e^{i(\mathbf{k}-\mathbf{p})\cdot\mathbf{r}}. \tag{10}$$

The angular shell integral is

$$\int \mathrm{d}\mu_k\, \delta(\epsilon - \omega_k) e^{i\mathbf{k}\cdot\mathbf{r}} = \frac{\sqrt{\epsilon^2 - m^2}}{4\pi^2} \operatorname{sinc}\Big(r\sqrt{\epsilon^2 - m^2}\Big). \tag{11}$$

Consequently,

$$\boxed{S_O(0, r) = \frac{1}{2\pi^3} \int_m^\infty \mathrm{d}\epsilon\, (\epsilon^2 - m^2) n(\epsilon)[1 + n(\epsilon)] \operatorname{sinc}^2\Big(r\sqrt{\epsilon^2 - m^2}\Big).} \tag{12}$$

Here $\operatorname{sinc} z = \sin z / z$ with $\operatorname{sinc} 0 = 1$. The zero-frequency symmetrized spectrum equals this expression; no additional factor of two is needed.

Independent momentum-transfer representation

Fourier transform Eq. (12) in space. Equal bath energies fix the longitudinal incoming momentum to $Q/2$, giving, for $Q > 0$,

$$\widetilde{S}_O(0, Q) = \frac{T}{2\pi Q} n\Big(\sqrt{m^2 + Q^2/4}\Big). \tag{13}$$

This is nonnegative, so all finite spatial covariance matrices are positive semidefinite. Its integrable $1/Q$ behavior for $m > 0$ does not justify substituting the exactly global, conserved $Q = 0$ coupling. Numerical Fourier inversion supplies an independent normalization check.

At coincident positions the finite thermal coefficient is

$$S_O(0, 0) = \frac{-mT^2 \ln(1 - e^{-m/T}) + T^3 \operatorname{Li}_2(e^{-m/T})}{\pi^3}. \tag{14}$$

It follows by using $n(1 + n) = -T\partial_\epsilon n$ and integrating by parts. In particular $[S_O] = \text{energy}^3$.

3 Visibility decay and the mixed-field extension

For a heavy probe with stationary paths, the local coupling is $H_{\rm int} = -c_a(P_L O_L + P_R O_R)$. Assume weak coupling, an effectively infinite stationary bath, and a controlled Markov approximation. The leading generator has the positive coefficient matrix $c_a^2 S_O(0, |\mathbf{x}_i - \mathbf{x}_j|)$:

$$\mathcal{L}\rho = c_a^2 \sum_{i,j=L,R} S_{ij} \left(P_j \rho P_i - \tfrac{1}{2}\{P_i P_j, \rho\} \right). \tag{15}$$

Its off-diagonal matrix element decays as

$$\gamma(r) = c_a^2 [S_O(0,0) - S_O(0,r)], \qquad \frac{V(\tau)}{V_{\rm reference}(\tau)} \simeq e^{-\gamma(r)\tau}. \tag{16}$$

The reference must remove this scalar channel while treating all ordinary instrument effects consistently. It is a theoretical reference here. The path populations remain unchanged under this pure-dephasing generator; the final recombination fringe follows the ordinary Born rule.

Independent collision check. In the heavy-target Born limit, $\sigma = c_a^2/\pi$ from Eq. (5). Multiplying the thermal number density by speed k/ω_k, Bose final-state enhancement, and isotropic path distinguishability gives

$$\gamma(r) = \frac{c_a^2}{2\pi^3} \int_0^\infty \mathrm{d}k \, \frac{k^3}{\omega_k} n_k(1+n_k)[1 - \mathrm{sinc}^2(kr)]. \tag{17}$$

Changing variable to $\epsilon = \omega_k$ reproduces Eq. (16). This connects the coefficient to conventional thermal collisional decoherence [3], rather than a modified Born rule.

Full quadratic coupling of two independent mass modes

For Eq. (6), with the modes at a common temperature, $\langle \delta B(x) \delta B(0) \rangle = 2 \sum_{ij} C_{ij}^2 G_i^>(x) G_j^>(x)$. Set $k_i(\epsilon) = \sqrt{\epsilon^2 - m_i^2}$ on its allowed shell. The same calculation gives

$$\boxed{\gamma_{\rm mixed}(r) = \frac{1}{2\pi^3} \sum_{i,j} C_{ij}^2 \int_{\max(m_i,m_j)}^\infty \mathrm{d}\epsilon \, k_i k_j n(\epsilon)[1+n(\epsilon)] \times [1 - \mathrm{sinc}(k_i r)\,\mathrm{sinc}(k_j r)].} \tag{18}$$

Both $i \neq j$ orderings are included. A mixed coupling can convert one species into the other at the same energy; this requires the threshold $\epsilon \geq \max(m_i, m_j)$. The fixed heavy probe absorbs momentum with negligible recoil energy. A finite-mass probe needs recoil kinematics.

Every integrand in Eq. (18) is nonnegative. If all masses coincide, the answer is proportional to $\mathrm{Tr}(C^2)$ and is invariant under orthogonal changes of the degenerate basis. For unequal masses, the cross-spectrum can have negative spatial values; the one-field bound $\gamma \leq S_B(0,0)$ must be replaced by $\gamma \leq 2S_B(0,0)$.

4 Limits that can falsify an implementation

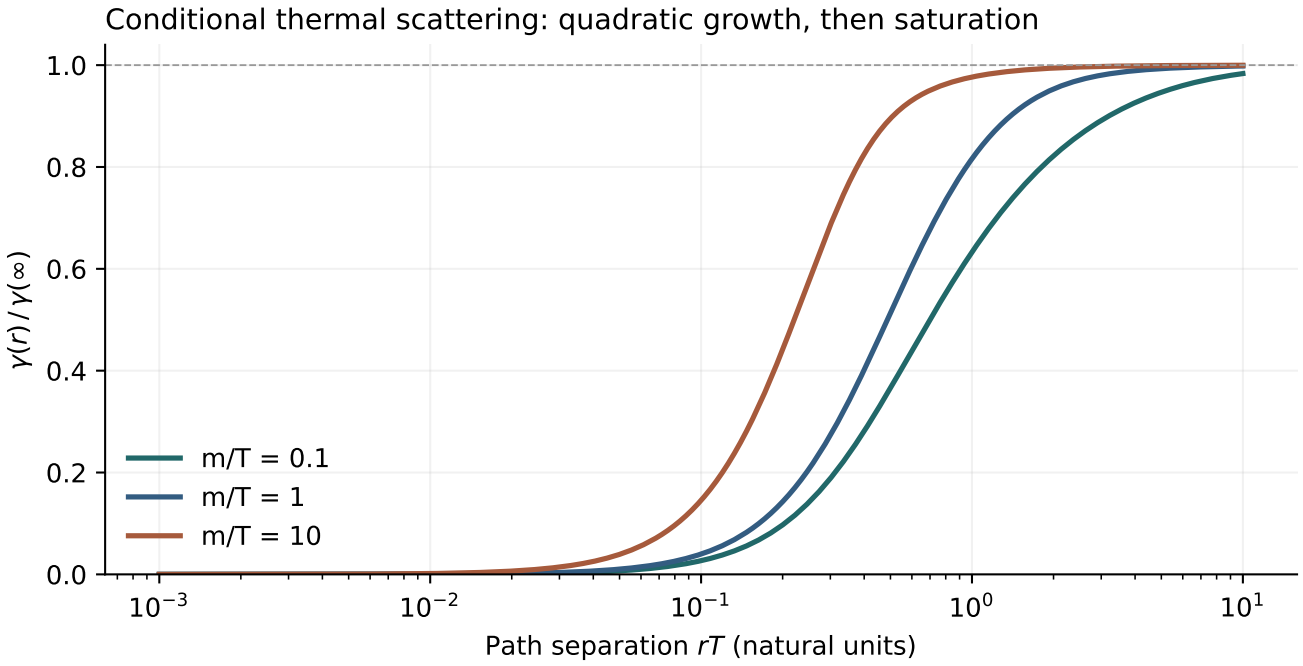
For one scalar, Eqs. (12)–(16) imply

$$0 \le \gamma(r) \le c_a^2 S_O(0,0), \qquad \gamma(0) = 0, \tag{19}$$

$$\gamma(r) = \frac{c_a^2 r^2}{6\pi^3} \int_m^\infty \mathrm{d}\epsilon\, (\epsilon^2 - m^2)^2 n(1+n) + O(r^4), \tag{20}$$

$$\gamma(\infty) = c_a^2 S_O(0,0), \qquad S_O(0,r) \sim \frac{Tn(m)}{4\pi^3 r^2} \quad (r \to \infty). \tag{21}$$

The small-distance expansion requires $kr \ll 1$ over the populated modes. It does not apply when the arms are already decorrelated.



Plot. Deterministic evaluation of the conditional formula, normalized separately by each curve's saturation rate. The curves are not observations; normalization hides the strong mass/temperature dependence of their amplitudes.

For $T \ll m$, the leading local spectrum is

$$S_O(0,0) \simeq \frac{(mT^2 + T^3)e^{-m/T}}{\pi^3}. \tag{22}$$

The suppression is $e^{-m/T}$, since one existing thermal particle scatters. For $r\sqrt{mT} \ll 1$ as well, $\gamma(r) \sim 4c_a^2 m^2 T^3 r^2 e^{-m/T}/(3\pi^3)$. As $m/T \to 0$, the local coefficient approaches $T^3/(6\pi)$; this limiting coefficient alone does not establish every massless Markov assumption.

Vacuum boundary. At $T = 0$ the zero-frequency rate vanishes for $m > 0$. Vacuum pair production starts at $2m$. This statement concerns the long-time leading rate and does not exclude switching transients, dressing, or finite-time entanglement with the vacuum.

Equal means are not a null. A homogeneous bath has $\langle O_L \rangle = \langle O_R \rangle$ and therefore no mean relative phase, but its difference noise is nonzero for $r > 0$. At $m/T = 1, rT = 1$, the calculated rate is about 0.8162 of its large-separation value. Exact cancellation requires identical coupled fluctuations, not just equal classical background values.

5 Using the same portal in collider and visibility calculations

An additional consistency relation follows without choosing a portal strength. For one real scalar with $2m < m_h$, the tree-level Higgs partial width in this convention [2] is

$$\Gamma_{h\to\phi\phi} = \frac{\kappa^2 v^2}{32\pi m_h}\beta, \qquad \beta = \sqrt{1-4m^2/m_h^2} > 0. \tag{23}$$

Eliminating κ^2 between this expression and Eq. (3) gives a conditional relation to the heavy-nucleon visibility rate:

$$\boxed{\gamma_N(r) = \frac{8\pi f_N^2 m_N^2}{v^2 m_h^3 \beta}\Gamma_{h\to\phi\phi}\,[S_O(0,0) - S_O(0,r)].} \tag{24}$$

This relation uses the same portal parameter on both sides. It avoids treating the interferometric rate as a freely adjustable coefficient once the scalar state and physical matching are specified.

If both daughters are invisible in the collider detector, a valid upper limit on the total invisible width also bounds this partial width. It can then be inserted into Eq. (24), conditionally on the scalar mass and bath spectrum. No current collider limit is inserted here and no new experimental exclusion is claimed. A branching-fraction limit also requires a consistent total-width model before conversion to a partial-width bound. At threshold $\beta = 0$ the division is invalid; near threshold the inferred bound becomes weak.

Degenerate mixed modes

When all included scalar mass modes have the same mass and thermal propagator, the inclusive sector width is

$$\Gamma_{h\to ss} = \frac{v^2\beta}{32\pi m_h}\,\mathrm{Tr}(K^2). \tag{25}$$

The diagonal channels carry the identical-particle factor; each distinct pair contributes twice a diagonal channel with the same portal entry. Since $\gamma = \mathrm{Tr}(C^2)\Delta S_O$, the same relation (24) holds with this inclusive sector width. Applying an invisible-width bound requires every included daughter mode to be invisible in the detector. Mass degeneracy alone is insufficient when populations, interactions, or thermal mass corrections distinguish the modes. Nondegenerate modes require their own phase-space factors and Eq. (18).

Optional naturalness restriction, explicitly conditional

If one further imposes $m^2 = m_0^2 + \kappa v^2/2$ with $m_0^2 \geq 0$ and $\kappa \geq 0$, then

$$\kappa \leq \frac{2m^2}{v^2}, \qquad \gamma_N(r) \leq \frac{f_N^2 m_N^2 m^4}{v^4 m_h^4}\,[S_O(0,0) - S_O(0,r)]. \tag{26}$$

This is a no-cancellation prior for the stated mass decomposition, not a bound implied by symmetry or a proof of loop-level naturalness. A protected or tuned construction may not obey it. Running couplings, thresholds, Higgs mixing, and quantum corrections require consistent matching before the tree-level relations can support a precision comparison.

Remaining empirical obstruction. A collider measurement constrains interactions, not the local occupation of an unobserved scalar bath. Without independent state information, Eq. (24) cannot assign a numerical laboratory visibility signal.

6 Validity, unresolved inputs, and experimental meaning

The underlying scalar state is Gaussian; the quadratic operator is not. Equation (9) fixes its two-point function but does not eliminate higher connected cumulants. Exponential visibility is a leading weak-coupling Markov prediction, not an exact finite-coupling solution. Weak-coupling limits require technical hypotheses [4]; they are not proved for every unbounded composite-field model by writing a Lindblad equation.

Regulation. Normal ordering removes the vacuum mean, not every coincident-point divergence. Use a smooth regulator on each field mode, for example $F_\Lambda(k) = e^{-k^2/(2\Lambda^2)}$, before making the full dynamical calculation. The one-scalar thermal integrand receives $F_\Lambda(k)^4$ and has a finite limit as $\Lambda \to \infty$. This is a statement about the displayed thermal coefficient; it does not renormalize all Lamb shifts or higher orders. Related scalar open-system calculations exhibit finite-time renormalization issues [5], and quadratic detectors have additional ultraviolet subtleties [6]. A physical EFT cutoff must remain below its breakdown scale; taking the convergent thermal integral to infinity is justified only when its relevant modes lie well below it.

Scale and state requirements. Higgs matching requires relevant momentum transfers small compared with m_h. Probe recoil, appreciable scalar interactions, spatially varying populations, field-dependent detector response, and bath backreaction require further work. A bath at temperature T is a stipulated state, not evidence of thermalization or a cosmological abundance. Take the continuum bath limit before the long-time limit; a finite free box has recurrences. The coupling must remain weak on the regulated bath's correlation timescale.

What an experiment could constrain. With a measured or independently justified scalar state and an apparatus transfer model, visibility data can constrain Eq. (16) or Eq. (18). Without such state information, an absent signal cannot bound the portal coupling independently of scalar abundance: the bath can simply be unpopulated. Even a nonzero decoherence rate does not uniquely identify a scalar mechanism. The massless limiting amplitude $c_a^2 T^3/(6\pi)$ makes an amplitude-only coupling/temperature degeneracy explicit.

Required measurement model. The next empirical step needs a named matter-wave probe, physical path profiles and duration, bath-state bounds, the associated coupling/form factor, a visibility likelihood including ordinary scattering and technical noise, and an independently calibrated reference. This note does not supply a photon Mach–Zehnder response by substituting nucleon parameters. No laboratory data were collected and no discovery is inferred.

Relation to earlier mechanisms. This is an equilibrium thermal quadratic-portal branch. It neither derives nor assumes $\gamma_k(E) = \gamma_{k0} e^{\eta_k E}$, a hidden S^2E source, or the interpretive identification of fields with consciousness or ethics. Its role is to test a physical interaction already present in the corpus, not to declare all proposed mechanisms equivalent.

7 Verification record and corrections

`check_quadratic_bridge.py` evaluates the derived expressions and writes `checks.json`. The final script hash, versions, and numerical results are recorded there. All calculations are theoretical; no laboratory or simulated detection data are used.

Check	Scope
Energy integral vs dilogarithm	Local thermal coefficient and factors of π.
Spatial Fourier inversion	Independent momentum-transfer representation.
Scattering flux vs noise rate	Portal vertex and visibility-rate normalization.
Finer grid and longer integration range	Numerical resolution and thermal tail.
Small separation and low/high temperature	Limiting formulas and domain checks.
Eight-position covariance matrix	Numerical positivity check in a nontrivial geometry.
Regulated thermal rates	Convergence of this coefficient only.
Mixed masses and degenerate rotation	Thresholds, off-diagonal counting, basis invariance.

The first coarse energy grid missed rapid oscillations in one corner of the test range. Increasing its resolution removed a relative discrepancy of about 6.8×10^{-6}. The comparison is retained as a reason for the resolution check, rather than treating the first passing-looking values as final. The numerical tests support the stated formulas; they are not a proof of their application to an actual device.

Two corrections to preserve alongside the historical note

The corrected 11 September reservoir note [7] still introduces a $\lambda(E_L - E_R)P_L \otimes B$ gate while describing a two-monitor repair. This is an additional differential-controller model; it does not follow from two local portal interactions. Physical access to both backgrounds needs causal transport or an explicitly justified quasi-static approximation. Equal average backgrounds do not establish its proposed null for the local quadratic interaction derived here.

That note also associates the spatial quadratic law with $\ell_c \ll \Delta x$. In a smooth correlated bath, the Taylor regime is $\Delta x \ll \ell_c$ for the relevant modes. Large separation yields saturation in the one-scalar model. These findings are recorded as an addendum; the historical source has not been silently rewritten.

Current claim boundary

Portal matching, the free thermal spectral integrals, and the leading conditional visibility law have explicit derivations and computational cross-checks. Finite-time renormalized dynamics, measured bath inputs, device calibration, global parameter compatibility, quantum gravity, high-energy completion, and independent experimental replication remain open. Internal AI cross-checks are not external scientific verification.

8 Public experimental data located for the next test

The user reported no apparatus and asked that existing data be found. The following public data were located on 11 October 2026. Their availability does not establish the proposed scalar interaction. The derivation and numerical checks above do not fit these measurements.

Lattice atom interferometry. Panda et al., *Coherence limits in lattice atom interferometry at the one-minute scale*, Nature Physics (2024), doi:10.1038/s41567-024-02518-9. The paper links an open figure-data deposit. Both its original version and the newer Zenodo record 15227841 were retrieved; all five files in each version passed their published MD5 checksums. The newer files include contrast versus hold time and measured fringes. These are published figure data, not a complete record of every experimental event or systematic uncertainty. The old and new files are not interchangeable: both figure numbering and values change.

A data-quality issue to resolve before fitting. The newer Figure 3a file has nine rows. Its last two rows have identical contrast and uncertainty, at very slightly different times near 70.071 s. They may represent the same measurement. They must not silently be counted as two independent observations. The record does not supply a full covariance matrix or a scalar-free visibility reference. The paper identifies ordinary apparatus dephasing, so observed contrast loss cannot be assigned to a new field by subtraction without an apparatus model.

Sodium-cluster matter-wave interferometry. Pedalino et al., *Probing quantum mechanics with nanoparticle matter-wave interferometry*, Nature 649, 866–870 (2026), doi:10.1038/s41586-025-09917-9. The linked Zenodo record 17502163 contains measurement scans, mass spectra, a settings workbook, and analysis code. Its 88,626-byte archive was downloaded and its published MD5 checksum verified. This is a candidate for an apparatus-specific visibility analysis. The archive's code was inspected as source and was not executed. The published agreement with ordinary quantum interference does not establish a new scalar bath. Both Zenodo datasets carry CC BY 4.0 licenses.

Collider complement. The ATLAS HIGG-2021-05 public result supplies invisible-Higgs branching-fraction constraints and likelihood-scan plots. These can constrain the portal under the experimental production and decay assumptions. A plotted likelihood is not automatically a downloaded numerical likelihood, and an exclusion limit is not a detection. A numerical reinterpretation is deferred until its input and width assumptions have been checked.

Local evidence and next step. A local magnetometer CSV matches the recovered pilot's recorded hash, but the accompanying account describes uncontrolled phone orientation and motion. Existing project interferometry fixtures are labeled synthetic. Neither is a measured test of Eq. (16). The next defensible step is to reproduce a public interferometer's baseline analysis, resolve data provenance issues, and derive its finite-path and composite-probe response before adding the scalar channel. Bath-state information remains necessary. The companion data catalog records locations, checksums, and these limits.

Sources and reproducibility

1. C. M. Baird / ZoraASI research collection, supplied four-part edition. Volume 4, physical PDF pp. 1706–1711: *A Two-Singlet Effective Theory and Its Experimental Interfaces*, 21 September 2026, v0.2. Volume 1, pp. 248–255: *A Conditional Mathematical Scaffold*. These are source proposals and audits, not independent validation.

2. P. Athron et al., *Status of the scalar singlet dark matter model*, Eur. Phys. J. C 77, 568 (2017), especially Eqs. 1 and 8. doi:10.1140/epjc/s10052-017-5113-1. Used for the conventional portal normalization and scattering cross-check; its historical parameter constraints are not imported as current bounds.

3. K. Hornberger and J. E. Sipe, *Collisional decoherence reexamined*, Phys. Rev. A 68, 012105 (2003). arXiv:quant-ph/0303094. Context for scattering-induced path decoherence and normalization discipline.

4. E. B. Davies, *Markovian master equations*, Commun. Math. Phys. 39, 91–110 (1974). Publisher record. Context for conditional weak-coupling limits; not a claim that every hypothesis has been established for the full interacting scalar apparatus.

5. C. Burrage, C. Käding, P. Millington and J. Minář, *Open quantum dynamics induced by light scalar fields*. arXiv:1812.08760. Related Higgs-portal open dynamics and finite-time renormalization.

6. A. Sachs, R. B. Mann and E. Martín-Martínez, *Entanglement harvesting and divergences in quadratic Unruh–DeWitt detectors pairs*, Phys. Rev. D 96, 085012 (2017). arXiv:1704.08263. Different detector setting illustrating why composite-coupling UV control cannot be inferred from normal ordering alone.

7. *Reservoir derivation and fail-closed interferometric protocol for the MQGT–SCF rate-modulated GKSL sector*, corrected edition, 11 September 2026. Local source in `work/gate1-public/`: `Reservoir_Protocol_Note_2026-09-11_CORRECTED.tex`, especially lines 305–359 and 477–481 in the inspected snapshot.

Reproduction. In an environment with Python, NumPy, mpmath and Matplotlib, run `python check_quadratic_bridge.py -plot` in the research directory. Build this note with two passes of `pdflatex quadratic_portal_measurement_bridge.tex`. The separate JSON manifest records source and output hashes.

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